HONGYI LUO

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EDUCATION

Bachelor of Science in Computer Science and Music — **Sep 2021 - Expected Graduation: May 2025**

University of Illinois Urbana-Champaign — GPA: 3.86/4.00

Honors: On the Dean's List

COURSEWORKS

Software Engineering	Java, C, C++, Python, Computer Systems, Software Engineering Lab
Computation Theory	Discrete Structures, Data Structures, Algorithms, Programming Languages & Compilers, Models of Computation
Mathematics	Matrix Theory, Probability & Statistics for Computer Science
Applications	Game Development, Electronic Music Techniques, Music Computation
Future Courses	Database Systems, Data Mining, Artificial Intelligence, Machine Learning, Mobile Computing Application

TECHNICAL SKILLS and LANGUAGES

Programming Languages	Java, C, C++, Python, OCaml
Frameworks and Tools	LangChain/LangGraph, LLM, OpenAI, Anthropic, Alibaba Qwen, SpringBoot, Git, SQL, VS Code, IntelliJ IDEA, Android Studio, Unreal Engine 5
Spoken Languages	English (proficient), Mandarin (native) , Cantonese (native)

WORK EXPERIENCES

Jun 2024 – Aug 2024 **Software Developer | Intern**

AI Research and Development Dept., China Telecom See-Link Technology Co., Ltd., Guangzhou, China

- Conducted research on 5+ multi-agent frameworks and products, culminating in a comprehensive report
- Developed object recognition capabilities for video analysis, achieving 100% accuracy in identifying key objects
- Maintained and debugged AI management platform code, resolving 10+ user-reported issues

Jun 2023 – Aug 2023 **Software Developer | Intern**

Software Development Dept., Unicorn Technology Development Co., Ltd., Guangzhou, China

- Contributed to Hitachi Elevator Monitoring System for 1 central and 3 regional centers in China
- Developed 7 backend data interfaces integrating the Monitoring System with the Maintenance System

Aug 2022 – Sep 2022 **Software Tester | Part-time**

School of Music, University of Illinois Urbana-Champaign, Champaign IL, USA

- Conducted comprehensive testing of Harmonia© software and 2 associated websites, covering 100% of features
- Identified and reported 10+ bugs, collaborating with developers to verify fixes, reducing post-release issues by 15%

PROJECT HIGHLIGHTS

Jun 2024 **Object Recognition**

Large Language Models (LLM) · Prompt Engineering · Multi-agent · LangChain · LangGraph · OpenAI · Python

- Developed a multi-agent application integrating LangChain, LangGraph, and large language models (LLM)
- Implemented autonomous planning for video asset retrieval based on investigation tasks
- Created object recognition capabilities for video frame analysis
- Designed self-evaluation and adaptive planning features for recognition quality improvement

Feb 2024 – Mar 2024 **Island Adventure**

Unreal Engine 5 · Game Design · Game Development · Game Programming

- Developed an open-world adventure game using Unreal Engine 5 with Blueprints
- Designed diverse terrains, AI-driven enemies, and immersive elements to elevate player experience
- https://github.com/Luo-hongyi/Island-Adventure

Jun 2023 **AI Poet**

Large Language Models (LLM) · Prompt Engineering · Web Development · Web Services · Python

- Developed an AI-driven Discord chatbot integrated with large language models (LLM)
- Implemented prompt engineering techniques using Chinese poetry elements to provide emotional support
- Created real-time user interaction capabilities for understanding and responding to input
- https://github.com/Luo-hongyi/AI-Poet

Oct 2022 – Dec 2022 **Sky Route (Course Project)**

C++ · Algorithms · Data Analysis

- Developed an analysis program using OpenFlights.org data to identify key hub airports in specific regions
- Implemented algorithms to calculate airport betweenness centrality, utilizing Dijkstra's and BFS algorithms
- Applied software development lifecycle (SDLC) principles throughout the project
- https://github.com/Luo-hongyi/Sky-Route

EXTRACURRICULAR ACTIVITIES

Oct 2021 – Present **Leader, Plan P Music Studio**

An officially registered society at the University at Illinois Urbana-Champaign

- Demonstrated leadership by managing the university society, coordinating diverse projects and events
- Applied project management skills to oversee music arrangement and performance preparation
- Developed mentorship programs to enhance overall studio quality

Aug 2021 – Present **Music Arranger and Trombonist, Jasmine Field Orchestra**

Chinese Students and Scholars Association at the University at Illinois Urbana-Champaign

- Created musical arrangements to enhance individual and group performances
- Performed as trombonist in orchestra concerts and events